Sampling-boundary distortion in Sharpe ratio evidence from financial panels

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Abstract

Sharpe ratios computed from financial panels may appear overly precise if each asset-date return is treated as independent, even though portfolio performance is realized once per date. This paper studies that row-to-date sampling-boundary distortion using public Kenneth French panels, AQR factor series, and Monte Carlo designs. It introduces Sharpe UVIF (Unified Variance Inflation Factor), a diagnostic ratio of row-pooled IID Sharpe variance to date-level long-run Sharpe variance. Empirical claims are evaluated after date aggregation using heteroskedasticity-and-autocorrelation consistent (HAC) delta tests and moving-block or stationary bootstraps; menu, same-date placebo, and turnover-cost diagnostics are reported separately. When the true edge is zero under dependent null designs, row-naive tests reject about 35 percent at nominal 5 percent, while date-level corrections stay much closer. Canonical momentum survives; weaker panel claims do not.

Keywords: Sharpe ratio; panel data; cross-sectional dependence; serial dependence; HAC/bootstrap inference; portfolio evaluation
JEL Codes: C12; C15; G11; G12; G17

1 Introduction

1.1 Problem: Row-to-date sampling boundary

Empirical finance often tests signals in entity-time panels: factor sorts, anomaly screens, automated portfolio-signal evaluations, model-selection grids, and trading-strategy backtests. A screen selects assets on each date, forms a portfolio, and reports a Sharpe ratio. The statistical question is whether that Sharpe evidence is attached to the right sampling unit: selected asset-date rows or date-level portfolio returns.

The distinction matters because portfolio performance is realized once per date. If a researcher treats each selected asset-date return as an independent observation, the apparent sample size becomes the number of selected rows. If same-date rows share common shocks, however, the additional rows do not contain that much independent information. This row-to-date mismatch is the sampling-boundary distortion studied here. In the Monte Carlo null designs below, row-naive tests reject around 35 percent of the time at a nominal 5 percent level when the true edge is zero, while date-level HAC and dependent-bootstrap methods stay much closer to nominal size.

The underlying entity-time panel is denoted by $\mathcal{P} = \{(t, i, s_{t,i}, r_{t+1,i})\}$, where t indexes dates, i indexes assets or portfolios, $s_{t,i}$ is the signal, and $r_{t+1,i}$ is the next-period return. One evaluation failure mode is to evaluate a positive edge using all selected asset-date rows as though they were independent. This turns a panel with T dates and n_t selected rows per date into a nominal sample of $\sum_t n_t$, even when same-date rows share common shocks. The row-pooled construction is retained only as a diagnostic baseline against which date-level inference is compared.

The issue extends beyond a single trading-strategy application. Factor screens, anomaly panels, model-selection procedures, and automated portfolio-signal evaluations can all multiply observations across assets on the same date while the economic payoff is realized through a date-level portfolio return. The paper uses public panel and factor examples because they make the sampling boundary visible, but the statistical problem is broader: what uncertainty should be attached to a Sharpe estimate after cross-sectional row multiplication and serial dependence in the date-return series are recognized. Cross-sectional dependence determines how much extra information the additional selected rows actually contain, and serial dependence determines the long-run variance of the date series. Standard clustering techniques, including the finance-panel guidance of Petersen (2009), address uncertainty for linear means or regression coefficients; they do not by themselves map row-pooled Sharpe evidence to the long-run variance of the economically relevant date-level Sharpe functional.

Automated empirical signal screens can evaluate large numbers of entity-date rows and many menu variants in a single run. The distortion is largest when many same-date entities are selected, common shocks are strong, menus are searched, and the confirmatory statistic is a nonlinear Sharpe functional.

The intuition is simple. If 50 selected rows are generated from only a small number of economically relevant dates, the row count can make the Sharpe estimate look more precise than the portfolio-return evidence supports. The relevant sampling object is the date-level return series, not the multiplied row count. Sharpe UVIF records how far the row-pooled IID uncertainty sits from the date-level long-run uncertainty.

The required diagnostic is a ratio between the uncertainty attached to the date-level Sharpe claim and the uncertainty a row-pooled IID analysis would report. This ratio is the Sharpe Unified Variance Inflation Factor, or Sharpe UVIF:

$$\mathrm{UVIF}_{\mathrm{SR}} = \frac{\mathbb{V}\mathrm{ar}_{LR}(\widehat{\mathrm{SR}}_{date})}{\mathbb{V}\mathrm{ar}_{IID,row}(\widehat{\mathrm{SR}}_{row})}.$$

It is best interpreted as a sampling-boundary distortion ratio for reported Sharpe uncertainty. The numerator is the uncertainty of the economically relevant date-level Sharpe claim. The denominator is the uncertainty that would be reported by a row-pooled IID analysis of the selected entity-date returns. When the row-pooled and date-level estimators target the same Sharpe functional under exchangeability and equal-weight aggregation, Sharpe UVIF reduces to a design-effect variance inflation factor for the nonlinear Sharpe ratio. Sharpe UVIF is interpreted as a reported-uncertainty ratio between row-pooled IID Sharpe inference and date-level long-run Sharpe inference.

1.2 Contribution and empirical approach

This paper contributes to financial econometrics in three ways. First, it formalizes row-to-date sampling-boundary distortion for Sharpe-ratio evidence in financial panels. Second, it defines Sharpe UVIF as a diagnostic ratio between row-pooled IID uncertainty and date-level long-run uncertainty, extending standard design-effect logic to the nonlinear Sharpe functional. Third, public Kenneth French panels, AQR factor series, and Monte Carlo designs show that the correction preserves canonical WML momentum evidence while weakening weaker panel claims, and that row-naive tests overreject under dependent zero-edge nulls.

The empirical section checks the implementation on canonical French WML, evaluates French panel stress cases, and reports AQR factor benchmarks as time-series-only evidence. The public Kenneth French and AQR evidence is based on documented public data sources and reproduction files. The conservative empirical question is whether apparently significant portfolio-signal claims owe their precision to row-level sampling and disappear after date aggregation, dependence adjustment, and researcher-menu correction. The replication code also runs a Monte Carlo study comparing row-naive testing with date-IID bootstrap, HAC-delta, moving-block bootstrap, stationary bootstrap, and joint resampling decisions.

2 Literature and motivation

This paper sits at the intersection of literatures that are often discussed separately. Survey and grouped-error design effects explain why row counts overstate information (Kish, 1965; Moulton, 1986). Large-panel econometrics studies cross-sectional dependence directly (Chudik and Pesaran, 2015), and finance-panel standard error work explains why cross-sectional and time clustering matter in empirical finance (Petersen, 2009). Those results are central background, but they mainly address means or regression coefficients rather than nonlinear portfolio-performance evidence after entity-date rows have been aggregated into a date-level return. They do not define a target-aware distortion ratio for nonlinear portfolio performance statistics.

Sharpe-ratio inference follows a separate time-series literature. Lo (2002) studies Sharpe-ratio sampling under return dynamics, and robust Sharpe testing builds on HAC and bootstrap methods (Ledoit and Wolf, 2008; Opdyke, 2007; Andrews, 1991; Kiefer and Vogelsang, 2005; Kunsch, 1989; Politis and Romano, 1994; Lahiri, 2003). These tools define the date-level sampling law once the return series is specified. They do not quantify how much apparent Sharpe precision was created before the evidence was moved from selected entity-date rows to the economically relevant date-return boundary. The boundary problem occurs before Sharpe inference: which return series is the object? What is missing in the Sharpe-inference literature is an explicit mapping from row-pooled Sharpe evidence to date-level long-run Sharpe evidence.

Researcher-menu and factor-zoo work addresses a third source of false precision. White Reality Check and Romano-Wolf procedures control data-snooping and familywise error in menus (White, 2000; Romano and Wolf, 2005), while the empirical asset-pricing literature documents false discoveries, factor-alpha fragility, anomaly decay, and trading costs (Harvey et al., 2016; Harvey and Liu, 2020a,b; Bailey and Lopez de Prado, 2014; Holm, 1979; Fama and French, 1993; Carhart, 1997; Fama and French, 2010; McLean and Pontiff, 2016; Almgren and Chriss, 2001; Hasbrouck, 2009; Frazzini et al., 2015). This paper is adjacent to that literature but distinct: its object is the sampling unit and return boundary before a row-level Sharpe claim is interpreted as portfolio-performance evidence. It separates three failure channels that are often bundled together in practice: cross-sectional row multiplication, serial dependence in the date-return series, and researcher-menu search. Menu corrections control search across candidate strategies; they do not correct a mismatch between the row-level sampling unit used to report Sharpe evidence and the date-level return series that defines portfolio performance.

Recent empirical asset-pricing work asks whether published predictors survive stricter implementation, data, and multiple-testing standards. The present paper asks an upstream question: before a predictor is interpreted as portfolio-performance evidence, has the sampling unit been moved from selected entity-date rows to the date-level return that can be interpreted as portfolio-performance evidence?

3 Data and empirical design

The empirical design uses public Kenneth French momentum and Size/book-to-market portfolio panels as row-level applications, and public AQR factor series as pre-aggregated time-series benchmarks. The French row-level panels allow same-date row aggregation, selected-count diagnostics, and same-date placebo checks. The AQR series are useful factor-return benchmarks, but they do not contain constituent signal-return rows and are therefore reported as time-series evidence only. These public applications are benchmark illustrations of the method, not a survey of all asset-pricing anomalies.

The registry records all source URLs, source frequencies, annualization factors, and sample windows. The two French momentum-decile daily panels run from 1926-11-03 to 2026-04-29 with 26,131 active dates. The French 25 Size/BM stress case uses the trailing daily window 2012-06-29 to 2026-04-29 with 3,477 active dates. The AQR daily factor-series benchmarks end on 2026-03-31, with start dates determined by the public source series, and the monthly AQR value-and-momentum series runs from 1972-01-31 to 2026-03-31 with annualization factor 12. Daily sources use annualization factor 252.

The method is a dependence-aware inference design rather than a single test. Its first channel is cross-sectional: all evidence is translated from selected asset-date rows into one economically interpretable date-level portfolio return, so same-date clusters cannot create sample size by row multiplication. Its second channel is serial: all Sharpe decisions are made on the date-return series using HAC-delta and dependent-bootstrap procedures. For panel candidates with row-level signal-return pairs, the empirical implementation reports both channels before adding researcher-menu, placebo, factor-alpha, and turnover-cost sensitivity diagnostics.

Each panel candidate is reported at three levels: the row-naive baseline, the date-level confirmatory estimate, and the UVIF diagnostic gap between them. Figure 1 summarizes the workflow. The top row shows the maintained empirical path: selected entity-date rows are first aggregated to one return per date and then evaluated with date-return inference. The bottom row shows the diagnostic path: the row-pooled IID report is retained only to measure the sampling-boundary gap and to organize supplementary checks.

3.1 Date aggregation as the target boundary

Let S_t denote the selected entities on date t, let $Y_{t+1,i}$ be the signed one-period row return, and let $w_{t,i}$ be the economic weight assigned to selected row i. The portfolio return used

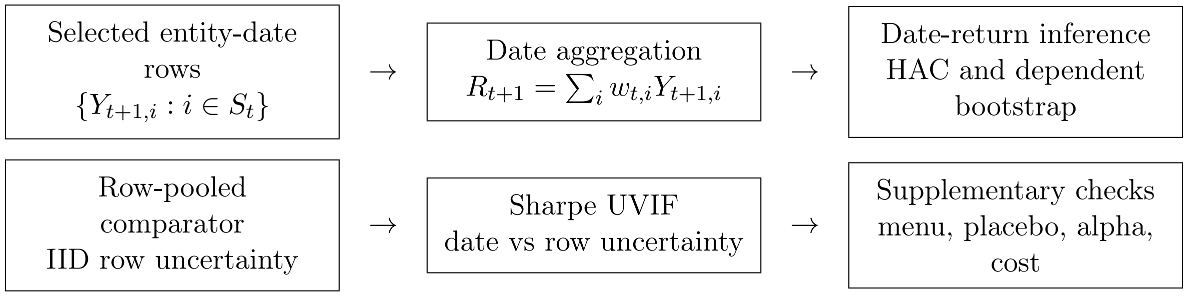

Figure 1. Inference workflow. The empirical claim is evaluated at the date-level portfolio-return boundary; row-pooled evidence is retained as a comparator rather than as the maintained sampling law.

for Sharpe inference is the date-level object

$$R_{t+1} = \sum_{i \in S_t} w_{t,i} Y_{t+1,i}, \qquad \sum_{i \in S_t} |w_{t,i}| = 1 \text{ on active public-panel dates.} \tag{1}$$

This is the target boundary. A row-pooled analysis estimates uncertainty as though the selected $Y_{t+1,i}$ were independent observations, while the confirmatory analysis estimates uncertainty for the realized date-return series $\{R_t\}$. In the empirical implementation, weights are normalized to the declared gross-exposure convention before returns are computed. Unless otherwise stated, rendered public panel tables use unit gross exposure within the selected signed-return set. For signed long-short rows, the selected positive and negative signs are combined into one unit-gross portfolio on each active date; dates with no selected row receive no active portfolio return. Sharpe calculations use active dates only unless otherwise stated.

In the public equal-weight threshold implementation, optional non-negative weights $a_{t,i}$ are normalized within the selected set $\mathcal{S}_t$, where $\mathcal{S}_t$ is produced by the declared signal and threshold rule, giving

$$\bar{r}_{t+1}(L, q, \tau) = \frac{\sum_{i \in \mathcal{S}_t} a_{t,i} y_{t+1,i}^{(L,q)}}{\sum_{i \in \mathcal{S}_t} a_{t,i}}. \tag{2}$$

The denominator is the total active economic weight on date t, not a generic row count. Under the primary equal-weighted evaluation $a_{t,i} = 1$, it reduces to the number of selected rows. Confidence-weighted and signal-magnitude-weighted returns are sensitivity checks.

3.2 Sharpe inference

The Sharpe ratio throughout is annualized mean excess return divided by period return volatility, with the source-specific annualization factor applied after the period mean and variance are computed. The inference design separates estimation from testing. It reports two-sided 95 percent intervals for $\widehat{\mathrm{SR}}$ and one-sided positive-edge p-values for $H_0 : \mathrm{SR} \leq 0$.

For a fixed return series with positive volatility, this is equivalent to a nonpositive mean excess return null. Bootstrap inference resamples the date-return series by IID date, moving block, or stationary bootstrap. The date-IID bootstrap is reported only as a comparator; dependence-aware conclusions rely on HAC, moving-block, and stationary-bootstrap procedures. The positive-edge bootstrap p-value uses one-draw smoothing:

$$p^{+} = \frac{1 + \#\{\widehat{\mathrm{SR}}_b^* \leq 0\}}{B+1}. \tag{3}$$

For HAC inference, reindex the finite realized date-return observations as R_u, $u = 1, \dots, T_R$, where R_u is the strategy excess return for the decision date after subtracting the applicable risk-free rate when the source is not already a long-short excess-return series. Let A_f denote the source-specific annualization factor: $A_f = 252$ for daily sources and $A_f = 12$ for monthly sources in the public configuration. Let $g_u = (R_u, R_u^2)'$, $\hat{\mu} = T_R^{-1} \sum_u R_u$, $\hat{m}_2 = T_R^{-1} \sum_u R_u^2$, and $\hat{v} = \hat{m}_2 - \hat{\mu}^2$. The denominator uses the T_R-moment variance, not the $T-1$ bias correction, because it is the plug-in moment entering the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}}.$$

This convention only changes finite-sample scaling and is kept consistent across the plug-in variance calculations. Following Ledoit and Wolf (2008), the Sharpe ratio is treated as a smooth functional $f(\theta)$ of the moment vector $\theta = \mathbb{E}[g_t]$. On sets where $m_2 - \mu^2$ is bounded away from zero, this map is Hadamard differentiable. The asymptotic normality of $\sqrt{T_R}(\widehat{\mathrm{SR}} - \mathrm{SR})$ follows from the functional delta method provided the long-run covariance matrix Ω is consistently estimated by a kernel-based HAC estimator. Evaluated at $(\hat{\mu}, \hat{m}_2)$,

$$\widehat{\mathrm{SR}} = \sqrt{A_f} \frac{\hat{\mu}}{\sqrt{\hat{v}}}, \qquad \nabla f(\hat{\mu}, \hat{m}_2) = \left(\sqrt{A_f} \frac{\hat{m}_2}{\hat{v}^{3/2}}, -\frac{1}{2} \sqrt{A_f} \frac{\hat{\mu}}{\hat{v}^{3/2}} \right)'.$$

With $\widehat{\Omega}$ the Bartlett-kernel HAC estimate of the long-run covariance of g_u, the HAC-delta variance is

$$\widehat{\mathbb{V}\mathrm{ar}}(\widehat{\mathrm{SR}}) = \nabla f(\hat{\mu}, \hat{m}_2)' \widehat{\Omega} \nabla f(\hat{\mu}, \hat{m}_2) / T_R. \tag{4}$$

Explicitly, for centered moments $\tilde{g}_u = g_u - \bar{g}$,

$$\widehat{\Gamma}_k = T_R^{-1} \sum_{u=k+1}^{T_R} \tilde{g}_u \tilde{g}_{u-k}', \qquad \widehat{\Omega} = \widehat{\Gamma}_0 + \sum_{k=1}^{K_T} \left(1 - \frac{k}{K_T + 1}\right) \left(\widehat{\Gamma}_k + \widehat{\Gamma}_k'\right).$$

This is the standard Newey-West/Andrews Bartlett-kernel construction (Newey and West, 1987; Andrews, 1991) applied to the joint process (R_u, R_u^2), so the off-diagonal terms carry the long-run covariance between the sample mean and second moment into the Sharpe-ratio delta method. The bandwidth K_T is a tuning choice and may depend on T. The default HAC bandwidth uses the Andrews automatic Bartlett rule, with fixed-lag and prewhitened sensitivities reported in the appendix. Important empirical claims should be checked against reported nearby bandwidths and against the block-bootstrap intervals rather than relying on a single HAC lag choice. The technical appendix reports prewhitened HAC and fixed-fraction HAC sensitivity checks.

Because financial date returns can be heavy-tailed, HAC-delta evidence is reported alongside bootstrap p-values and intervals. When HAC and dependent-bootstrap conclusions disagree, the result is reported as robustness-fragile rather than summarized as a positive-edge finding.

4 Implementation and decision rules

4.1 Public-panel momentum implementation

For the public French panel illustrations, the empirical implementation uses a skipped-momentum signal, within-date percentile thresholds, and equal-weighted signed returns. The construction is intentionally simple: it supplies a transparent panel example for the sampling-boundary diagnostic rather than an optimized trading rule. Detailed formulas for the skipped signal, selected set, tie handling, and selected-count granularity are reported in the technical appendix and replication files. In brief, for lookback L and skip q, the signal compounds prior returns through date $t - q$, excluding the outcome return; selected rows then receive signed one-period returns according to the signal sign.

4.2 Researcher menus, placebos, and factor alpha

For a fixed menu $\mathcal{M}$ of lookbacks and thresholds, the primary joint adjustment is Romano-Wolf stepdown max-t resampling over $\{R_u(m) : m \in \mathcal{M}\}$. Holm-adjusted marginal bootstrap p-values and White's Reality Check are secondary diagnostics. The Deflated Sharpe Ratio diagnostic of Bailey et al. (2014) is reported as an additional selection-bias check. In the public implementation, the menu is the finite pre-specified set

$$\mathcal{M} = \{(c, \theta_c, \tau) : c \in \mathcal{C},\ \tau \in \mathcal{T}_c\},$$

where θ_c denotes the candidate's fixed signal definition. For panel candidates, $\mathcal{T}_c = \{0.0, 0.3, 0.5, 0.7, 0.9\}$ and any lookback and skip values are fixed before the confirmatory run rather than selected after seeing results. Pre-aggregated AQR factor-series candidates use the singleton threshold menu $\mathcal{T}_c = \{0.0\}$; because they do not contain row-level constituent signals, same-date panel permutation is not applicable to them. Real researcher menus are usually correlated rather than independent, so the procedure uses Romano-

Wolf stepdown to control familywise error while preserving the joint dependence among menu elements.

The fixed-menu condition is a confirmatory convention, not a claim that research is never iterative. In practice, exploratory work should be separated from the final evaluation by freezing the signal definitions, lookbacks, thresholds, filters, cost assumptions, and benchmark model before the confirmatory run. If discarded variants influenced the final choice, they belong in the researcher menu or the exercise should be reported as exploratory. A stronger design uses a locked holdout period or an independently refreshed sample after the menu is frozen.

The public placebo is a same-date permutation design. For each date, the vector of signals is randomly permuted across portfolios, preserving the date's cross-sectional signal distribution and missingness while breaking the signal-return pairing. The signed next-day returns, confidence ranks, selected counts, turnover, and date aggregation are then recomputed. Repeating this procedure gives a permutation null for the observed public momentum panel.

This placebo is a test under a within-date exchangeability hypothesis: under the null, the assignment of signals $s_{t,i}$ to next-period returns $r_{t+1,i}$ is exchangeable within each date t. If signals are structurally tied to size, sector, liquidity, country, exchange, beta, or other common-factor state variables, an unrestricted same-date permutation can break the wrong null. In those cases the analysis should permute within pre-specified blocks, residualize returns against the structural variables before permutation, or mark the placebo as not applicable. The permutation check is therefore a design-specific diagnostic rather than a universal validity guarantee. If the permutation null has near-zero dispersion, the placebo is treated as uninformative for that threshold rather than as calibrated evidence for or against predictability. Such a diagnostic is reported as N/A: it is not counted as a permutation failure, and it is not sufficient for a full panel-rule pass.

The factor-alpha test estimates

$$R_t = \alpha + \beta_M(MKT_t - RF_t) + \beta_S SMB_t + \beta_H HML_t + \beta_U MOM_t + \varepsilon_t, \tag{5}$$

using HAC standard errors. The null is $H_0 : \alpha \leq 0$. Including momentum is essential because the public signal is itself momentum-like. The factor-alpha regression uses frequency-matched public factor returns and the same HAC lag rule as the main Sharpe analysis. The alpha table is a diagnostic rather than the main test for canonical benchmark validation.

4.3 Turnover-scaled costs

For the equal-weighted threshold portfolio, define target weights

$$w_{t,i} = \frac{\operatorname{sign}(s_{t,i})\mathbf{1}\{i \in \mathcal{S}_t\}}{|\mathcal{S}_t|}, \qquad w_{t,i} = 0 \text{ if } |\mathcal{S}_t| = 0. \tag{6}$$

Daily turnover is the half-L^1 distance between adjacent target weight vectors,

$$TO_t = \frac{1}{2}\sum_i |w_{t,i} - w_{t-1,i}|, \qquad \widehat{TO} = T_{TO}^{-1}\sum_t TO_t, \tag{7}$$

where the average is over adjacent dates with at least one active side. For cost c per full rebalance in decimal units, the cost-adjusted Sharpe ratio is

$$\widehat{\mathrm{SR}}_{net}(c) = \frac{\hat{\mu} - \widehat{TO}\,c}{\hat{\sigma}}\sqrt{A_f}. \tag{8}$$

The cost term $\widehat{TO}c$ is subtracted from the period mean return before annualization; $\widehat{TO}$ is averaged at the source frequency, daily for the public panel sources used here. This is a first-order cost stress. It assumes linear cost per unit of turnover and holds the return volatility fixed after subtracting costs. These assumptions are deliberately simple and are used only as a screening stress. The resulting cost drag may understate implementation costs when liquidity demand is material. They are not a full execution model for high-turnover or illiquid strategies. Such applications require spread, latency, borrow, and position-size models before any implementation claim. This is a conservative cost-stress screen, not an execution model. The empirical tables also report the break-even cost in basis points, defined by $\hat{\mu} - \widehat{TO}c = 0$. The public sorted-portfolio examples do not contain average daily volume or order-size information, so they do not support an execution-scale claim.

4.4 Decision rule for reported panel candidates

The implementation reports component diagnostics, and the pre-specified summary rule is deliberately stricter than any single statistic. A candidate is reported as passing the full panel evaluation only if all applicable diagnostics are informative and it clears the pre-specified thresholds:

$$p_{\text{boot}} \le \alpha, \qquad p_{\text{perm}} \le \alpha, \qquad p_{\text{HAC}} \le \alpha, \qquad p_{\text{RW}} \le \alpha, \qquad \widehat{\mathrm{SR}}_{net}(c) > 0.$$

In the reported tables $\alpha = 0.05$. Here p_{boot} is the one-sided dependent-bootstrap Sharpe p-value, p_{perm} is the same-date signal-permutation p-value, p_{HAC} is the HAC-delta positive-edge p-value, and p_{RW} is the Romano-Wolf familywise adjusted p-value over the fixed researcher menu. The final condition requires the turnover-cost-adjusted Sharpe to remain positive at the specified cost stress c. If the same-date permutation null is degenerate or otherwise uninformative, the candidate is reported separately as passing or failing the remaining applicable checks; it is not classified as a full pass. This is a conservative reporting policy, not a uniformly most powerful test, an optimal weighted test, or a claim that the diagnostics exhaust all possible failure modes. The component diagnostics are reported so readers can apply less conservative rules. The same-date permutation check is a panel placebo: it is applicable only when row-level signals can be permuted within date. Pre-aggregated factor series can still be economically interesting and can pass HAC-delta or Romano-Wolf tests, but they do not contain the row-level signal panel needed for this placebo. Such cases are reported as not applicable for the permutation check rather than as passed panel evaluations.

The stages have different interpretations. The permutation diagnostic asks whether the signal-return pairing beats a same-date placebo. The HAC diagnostic asks whether the resulting date-return Sharpe is positive after serial dependence and heteroskedasticity are accounted for. The bootstrap diagnostic asks whether that conclusion survives resampling of the date-return series. The Romano-Wolf diagnostic asks whether the claim survives the fixed researcher menu. The cost diagnostic reports whether net Sharpe remains positive after turnover-scaled frictions. Factor-alpha, holdout, subperiod, and stationarity diagnostics are reported alongside these checks to prevent a single favorable statistic from becoming the claim.

5 Sampling-boundary methodology

5.1 Mean-return special case

Assumption 1 (Equicorrelated row model)**.** *For each date t, selected row returns satisfy*

$$y_{t,i} = \mu + \lambda_t + \epsilon_{t,i},$$

where λ_t is a common date shock, $\epsilon_{t,i}$ is idiosyncratic, λ_t is uncorrelated with $\epsilon_{t,i}$, $\mathbb{V}\mathrm{ar}(\lambda_t) = \sigma_\lambda^2$, and $\mathbb{V}\mathrm{ar}(\epsilon_{t,i}) = \sigma_\epsilon^2$. The selected count is $n_t = \bar{n}$ for the displayed derivation, and dates are independent.

Under this model,

$$\sigma^2 = \mathbb{V}\mathrm{ar}(y_{t,i}) = \sigma_\lambda^2 + \sigma_\epsilon^2, \qquad \rho = \mathbb{C}\mathrm{orr}(y_{t,i}, y_{t,j}) = \frac{\sigma_\lambda^2}{\sigma_\lambda^2 + \sigma_\epsilon^2} \quad (i \neq j).$$

Thus ρ is the intraclass correlation induced by the common date shock; economically, it is the fraction of row-return variance explained by shocks shared by same-date selections.

In empirical diagnostics, ρ is estimated after the candidate threshold is fixed. For each panel candidate, the selected signed row returns are aligned by date and asset or portfolio, and $\hat{\rho}$ is the average off-diagonal sample correlation among active row-return series with overlapping dates; $\bar{n}$ is the average selected count across active dates. When the selected set varies over time, this equicorrelation estimate is a descriptive compression of the same-date covariance structure, not an identifying assumption for the confirmatory HAC or bootstrap tests.

Under the equicorrelated row model, the corresponding mean-return design-effect calculation gives the ratio of the naive row-pooled reported standard error of the mean to the date-level standard error:

$$\frac{se_{row,naive}}{se_{date,true}} = \{1 + (\bar{n} - 1)\rho\}^{-1/2}.$$

For $\rho > 0$ and $\bar{n} > 1$, the naive row-level standard-error report is too small.

This calculation is not new as a sampling result; it is the Moulton-like special case of the paper's UVIF logic, written for signed trading-panel rows and date-level portfolio returns. To avoid conflict with the researcher menu notation $\mathcal{M}$, write the cross-sectional variance design effect as

$$\mathrm{UVIF}_{\mu,xs}(\bar{n}, \rho) = 1 + (\bar{n} - 1)\rho.$$

The square root is reported when the object is a standard-error or t-statistic adjustment:

$$\mathrm{UVIF}^{1/2}_{\mu,xs}(\bar{n}, \rho) = \sqrt{1 + (\bar{n} - 1)\rho}.$$

Thus UVIF denotes a variance ratio, while $\mathrm{UVIF}^{1/2}$ denotes the associated standard-error multiplier. This mean-return calculation is expository: it makes the scale of understatement visible before the paper turns to HAC and dependent-bootstrap decisions on the realised date-return series. The magnitude can be economically large. The relevant inflation factor for the true standard error relative to the naive row report is $\mathrm{UVIF}^{1/2}_{\mu,xs}(\bar{n}, \rho)$. With $\bar{n} = 50$, increasing same-date correlation from $\rho = 0.10$ to $\rho = 0.60$ raises this factor from 2.43 to 5.51. The inflation factor rises mechanically when selected counts and same-date common-shock correlation rise.

The proof is given in the technical appendix.

Remark 1 (Heterogeneous factor covariance)**.** *The equicorrelated model is a closed-form illustration, not a restriction imposed by the protocol. Let a same-date signed-return vector satisfy*

$$y_t = \mu\mathbf{1} + Bf_t + u_t,$$

where B contains asset-specific loadings, $\mathbb{V}\mathrm{ar}(f_t) = \Sigma_f$, and $\mathbb{V}\mathrm{ar}(u_t) = D_t$. Then the *same-date covariance matrix is*

$$\Sigma_t = B\Sigma_f B' + D_t,$$

with heterogeneous pairwise correlations whenever loadings or idiosyncratic variances differ. For any displayed portfolio weights w_t, the relevant one-date variance is $w_t'\Sigma_t w_t$, not the row-independent variance. The method avoids estimating the full Σ_t as a prerequisite for inference by moving the evidence to the realised date-return series and then using HAC or dependent bootstrap inference on that series. The design-effect calculation is therefore best read as the transparent special case that explains why row pooling fails.

5.2 Unified variance inflation for the Sharpe ratio

The mean-return design effect is useful but incomplete because the reported performance metric is usually the Sharpe ratio. The Sharpe ratio is a nonlinear functional of the first two moments, so its variance inflation depends on the covariance of R_t and R_t^2, not only on the variance of R_t. This motivates the Unified Variance Inflation Factor:

$$\mathrm{UVIF}_{\mathrm{SR}} = \frac{\mathbb{V}\mathrm{ar}_{LR}(\widehat{\mathrm{SR}}_{portfolio})}{\mathbb{V}\mathrm{ar}_{IID,row}(\widehat{\mathrm{SR}}_{row})}. \tag{9}$$

The numerator is the long-run delta-method variance at the date-return boundary. The denominator is the row-pooled delta-method variance that would be reported if the selected rows were treated as independent observations. Equation (9) is therefore a reported-uncertainty distortion ratio. It compares the uncertainty of the economically relevant date-level Sharpe claim with the uncertainty that would be reported under row-pooled IID analysis. Under exchangeability of the linearized Sharpe influence function, and when the row-pooled and date-level estimators target the same equal-weight Sharpe functional, the cross-sectional part of this variance ratio reduces to the same $1+(\bar{n}-1)\rho$ design effect shown above, with ρ applied to the linearized Sharpe influence rather than to raw returns. Outside that special case, Sharpe UVIF is target-aware: it measures the gap between the row-level report and the date-level claim that can be interpreted as portfolio-performance evidence.

Assumption 2 (Sharpe UVIF regularity)**.** *The date-level process $G_t = (R_t, R_t^2)'$ is stationary with finite $4+\delta$ moments, $v_R = m_{2,R} - \mu_R^2 \geq \eta > 0$, and finite long-run covariance Ω_R. The aggregation rule is fixed before evaluation. The displayed decomposition uses a constant selected-count approximation $\bar{n}$, and the row-pooled IID denominator is a counterfactual reporting benchmark rather than the maintained sampling law.*

Proposition 1 (Sharpe UVIF decomposition)**.** *Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. Define $\theta_R = (\mu_R, m_{2,R})' = \mathbb{E}[G_t]$, $v_R = m_{2,R} - \mu_R^2 > 0$. For* *source-specific annualization factor A_f, define*

$$d_R = \nabla f(\theta_R) = \left(\sqrt{A_f}\frac{m_{2,R}}{v_R^{3/2}}, -\frac{1}{2}\sqrt{A_f}\frac{\mu_R}{v_R^{3/2}} \right)'.$$

Let $\Gamma_k^R = \mathbb{C}\mathrm{ov}(G_t, G_{t-k})$ and $\Omega_R = \Gamma_0^R + \sum_{k\geq 1}(\Gamma_k^R + \Gamma_k^{R\prime})$. For the selected row return $Y_{t,i}$, let $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$, $\theta_Y = \mathbb{E}[H_{t,i}]$, $d_Y = \nabla f(\theta_Y)$, and $\Psi_Y = \mathbb{V}\mathrm{ar}(H_{t,i})$. With a constant selected count $\bar{n}$, the first-order Sharpe variance inflation relative to a row-pooled IID report is

$$\mathrm{UVIF}_{\mathrm{SR}} = \underbrace{\frac{\bar{n}\, d_R'\Gamma_0^R d_R}{d_Y'\Psi_Y d_Y}}_{\textit{cross-sectional UVIF component}} \times \underbrace{\frac{d_R'\Omega_R d_R}{d_R'\Gamma_0^R d_R}}_{\textit{Sharpe long-run serial factor}}. \tag{10}$$

Moreover, the serial factor can be written as

$$1 + 2\sum_{k=1}^{\infty}\psi_k^{\mathrm{SR}}, \qquad \psi_k^{\mathrm{SR}} = \frac{d_R'\Gamma_k^R d_R}{d_R'\Gamma_0^R d_R},$$

and the scalar long-run variance entering the numerator is

$$d_R'\Omega_R d_R = \frac{A_f}{v_R^3}\left[m_{2,R}^2\Omega_{11} - \frac{\mu_R m_{2,R}}{2}(\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4}\Omega_{22} \right], \tag{11}$$

where Ω_{ab} denotes the (a, b) element of Ω_R.

The proof is given in the technical appendix.

5.3 What Proposition 1 does and does not identify

The decomposition is a first-order variance calculation for a fixed aggregation rule and a constant selected-count approximation. It identifies how a row-pooled IID Sharpe uncertainty report changes when the claim is moved to the date-return boundary and when the date series has long-run serial covariance in (R_t, R_t^2). It does not identify a causal trading edge, does not prove that row and date Sharpe functionals are equal outside the stated exchangeability conditions, and does not replace the confirmatory date-level HAC, dependent-bootstrap, researcher-menu, placebo, factor-alpha, and cost diagnostics. The empirical role of Sharpe UVIF is therefore diagnostic: it explains the sampling-boundary distortion that the reported diagnostics then evaluate with date-level inferential procedures. The sample UVIF reported in the empirical section is a plug-in diagnostic for this population ratio; confirmatory rejection decisions do not depend on estimating UVIF consistently.

Remark 2 (Connection to Moulton-like design effects)**.** *If the linearized row Sharpe influence $\zeta_{t,i}^Y = d_Y'(H_{t,i} - \theta_Y)$ is exchangeable within date with intraclass correlation ρ_ζ,* *and the date-level linearization is the equal-weighted average of those row influences, the cross-sectional factor in* (10) *reduces to* $1 + (\bar{n} - 1)\rho_\zeta$. *The mean-return design effect is the special case in which the influence variable is* $Y_{t,i} - \mu$. *The Sharpe case replaces raw return correlation by correlation of the linearized nonlinear-performance influence, which is why skewness, kurtosis, and volatility clustering enter the variance inflation.*

Fixed-menu validity. The HAC-delta and block-bootstrap procedures are used under the standard conditions for a fixed finite researcher menu: stationary date-return series, adequate moments for the Sharpe delta-method expansion, non-degenerate volatility, and block lengths satisfying $\ell \to \infty$ with $\ell/T \to 0$. Under those conditions, standard delta-method and block-bootstrap arguments support marginal Sharpe inference for each fixed menu element, and Holm or Romano-Wolf stepdown procedures control familywise error for the fixed menu (Holm, 1979; Kunsch, 1989; Romano and Wolf, 2005; Politis and Romano, 1994; Lahiri, 2003). These standard conditions do not cover long memory, infinite-variance returns, growing menus, structural breaks, or nonstationary date returns. The empirical stationarity, holdout, placebo, factor-alpha, and cost outputs are therefore diagnostics rather than proof that every assumption holds in every market setting.

6 Synthetic positive-control calibration

The reporting rule is not mechanically anti-discovery. To check that the decision criteria can pass a true signal, a synthetic panel is generated with a persistent predictive signal, moderate same-date dependence, and serial dependence in the date-level returns. The signal is constructed before inference is run, the same date-aggregation and reporting criteria are applied, and the resulting positive-control candidate passes HAC, Romano-Wolf, and same-date permutation checks. This calibration is not used as evidence for a public trading strategy; it shows that the procedure can accept known positive evidence when the data-generating process contains a genuine date-level edge.

The positive-control DGP uses $T = 5000$ dates and $N = 100$ entities, annual alpha 0.12 (daily alpha 0.12/252), idiosyncratic volatility 0.01, common-factor loading 0.50, AR(1) common-factor coefficient 0.10, signal-stay probability 0.98, and seed 777. The detailed output is kept in the technical appendix because this is an illustrative sanity check, not empirical evidence for a public anomaly.

Appendix material. The worked numerical example, full proof details, positive-control data generation notes, and secondary robustness tables are reported in the technical appendix. The main manuscript keeps only the decision rule and the evidence needed to evaluate it.

7 Empirical results: benchmark illustrations

The public evidence is illustrative rather than a broad anomaly-panel survey. It evaluates the canonical Kenneth French winner-minus-loser momentum decile benchmark, a broader Kenneth French decile-rank threshold profile, Kenneth French 25 Size/BM portfolios with a skipped-momentum signal, and public AQR value, momentum, quality, beta, and HML-style factor candidates. The AQR series are economically meaningful factor-series benchmarks, but they are pre-aggregated series rather than row-level constituent panels; when the same-date permutation check cannot be applied, the benchmark is outside the full panel rule. The main tables therefore separate panel candidates from single-series factor benchmarks. Failed public-data access attempts are kept in the provenance files rather than treated as main empirical results. The empirical section is intentionally a public-data demonstration rather than an anomaly census; the contribution is the sampling-boundary diagnostic and Monte Carlo size evidence.

Table 1 reports the compact panel-candidate status summary, including Sharpe, HAC evidence, permutation status, net Sharpe, and the final interpretation. Table 2 compares the row-naive report, date-level HAC inference, dependent-bootstrap inference, and Sharpe UVIF for the same panel candidates. The technical appendix reports the full secondary table set: AQR time-series benchmarks, WML validation, horizon sensitivity, same-date permutation details, HAC bandwidth/prewhitening/fixed-b robustness, rule sensitivity, holdout splits, and cost sweeps. The reproducibility package records the fixed public sources, candidate types, source frequencies, annualization factors, thresholds, lookbacks, skips, eligibility rules, and configuration hashes used in the reported tables.

The factor-alpha, data-snooping, stationarity, holdout, and cost tables are secondary diagnostics, not replacements for the primary date-return decision. A raw positive Sharpe that does not survive FF3 plus momentum alpha, bootstrap robustness, Romano-Wolf/White checks, same-date placebos, or plausible costs should not be interpreted as reliable portfolio-performance evidence. This is the empirical implication of the sampling-boundary design: familiar public candidates can look attractive at one boundary and fail at the date-level decision boundary.

The main empirical tables report the row-level panel candidates only. Pre-aggregated AQR factor series, benchmark-validation details, horizon sensitivity, and full robustness diagnostics are reported in the technical appendix.

The comparator table makes the standard-practice contrast explicit: row-naive evidence is shown next to date-level HAC and bootstrap inference on the portfolio return series.

The canonical French WML candidate is expected to match the direct Kenneth French winner-minus-loser decile construction and is reported as a public benchmark check. Its

Table 1. Panel candidates and final evaluation status.

Panel candidate	SR	HAC p+	Perm p	Net SR	Status
French momentum deciles, rank-threshold panel	0.447	7.54e-05	N/A	0.447	Permutation uninformative; remaining criteria pass
French WML decile benchmark	0.497	5.53e-06	1.00e-04	0.497	Passes full evaluation
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.943	-0.050	Fails statistical and cost checks

Note: Permutation N/A means that the same-date placebo was uninformative for that threshold; it is not counted as a full pass.

Table 2. Comparison with common row and date-level inference choices for panel candidates.

Panel candidate	row p+	date-IID p+	date-HAC p+	block p+	UVIF
French momentum deciles, rank-threshold panel	1.70e-04	2.00e-04	7.54e-05	2.00e-04	1.000
French WML decile benchmark	4.85e-04	2.00e-04	5.53e-06	2.00e-04	1.000
French 25 Size/BM, skipped-momentum panel	0.253	0.417	0.412	0.422	9.002

Note: The row-naive column treats selected rows as independent. Date-IID, date-HAC, and block-bootstrap columns operate on the date-level portfolio return series.

full-rule pass shows that date-level inference can preserve canonical evidence when the signal is strong enough and the placebo, menu, dependence, and cost diagnostics do not overturn it. The broader rank-threshold profile and the French Size/BM dynamic-momentum panel are evaluated as stress cases rather than as canonical anomaly replications. The appendix reports that the Size/BM horizon sensitivity does not rescue the stress case: longer lookbacks reduce turnover but leave net Sharpe slightly negative, while empirical Sharpe UVIF remains about 11–13. Pre-aggregated AQR factor series are reported separately as time-series benchmarks because they do not provide row-level signal-return pairs for same-date permutation or row-boundary diagnostics.

8 Monte Carlo evidence: 35 percent row-naive rejection

The Monte Carlo study is designed as a size-and-power check of the sampling boundary, not as a search for a favorable strategy. Each replication generates a symbol-date return panel, applies the same date-aggregation rule used in the empirical analysis, and then asks whether each inference method rejects a positive-edge null at the nominal 5 percent level. The base panel is

$$r_{t,i} = \mu_t + \beta_i' f_t + \epsilon_{t,i}, \qquad f_t = \Phi f_{t-1} + u_t.$$

The grid varies serial dependence, cross-sectional dependence, GARCH volatility, multifactor exposure, and regime switching. The high-persistence AR(1) case is reported as such; it is not labeled as a long-memory design. The high-volatility GARCH case is not described as an infinite-variance heavy-tail design. The simulation uses 1000 Monte

Carlo replications for the displayed size table, $T = 1000$ dates and $N = 50$ entities in the main null designs, 5000 bootstrap draws per simulation, and seed-indexed replications generated from the simulation index. Moving-block and stationary bootstrap procedures use $\ell = \lceil T^{1/5} \max(D, 1) \rceil$, capped at $\lfloor T/3 \rfloor$, and the stationary bootstrap uses the same mean block length. The full DGP grid, seed policy, configuration hash, and run metadata are included in the reproducibility package. The null designs set the true edge to zero while varying cross-sectional correlation and serial memory; the power designs then increase the true annualized Sharpe on the same dependence-aware decision boundary. The important object is the gap between the row-naive rejection rate and the rejection rates of methods that operate on date-level returns. In the displayed null grid, the row-naive test overrejects around 35–37 percent under dependent nulls, while HAC-delta, moving-block bootstrap, stationary bootstrap, and Romano-Wolf remain much closer to nominal size. In the power grid, row-naive rejection reaches about 50 percent at a true annualized Sharpe near 0.19, before the dependence-aware methods show comparable power; that gap reflects null overrejection rather than valid incremental power.

The size and power experiment compares row-level naive testing, date-IID bootstrap, HAC-delta inference, moving-block bootstrap, stationary bootstrap, and Romano-Wolf joint resampling. The coverage experiment focuses on the date-level Sharpe target, so row-naive intervals are interpreted as coverage of the economically relevant date-level claim rather than coverage of a separate row-distribution Sharpe. The purpose is not to rank methods by power alone. A method that appears powerful because it overrejects under the null is not acceptable for confirmatory evaluation. Date aggregation removes the row-boundary error; HAC and dependent bootstrap address serial dependence. The date-IID bootstrap is therefore a useful comparator, but not the preferred dependent-return procedure.

Figure 2 and Table 3 give the main size evidence. The row-naive column is the familiar symbol-date analysis; the HAC-delta, moving-block, stationary, and Romano-Wolf columns make decisions from the date-return series. When row-naive rejection is high in null or near-null designs but the dependence-aware methods remain near nominal size, the apparent extra power is a false precision gain from treating same-date rows as independent evidence. The technical appendix reports the design contract, resampling settings, coverage table, target-boundary diagnostics, design sweeps, power table, and power-curve figure.

The null-size figure and table are the core simulation results: under dependent nulls, row-naive testing rejects about one third of the time at a nominal 5 percent level, while date-level HAC, block-bootstrap, stationary, and Romano-Wolf procedures stay much closer to size.

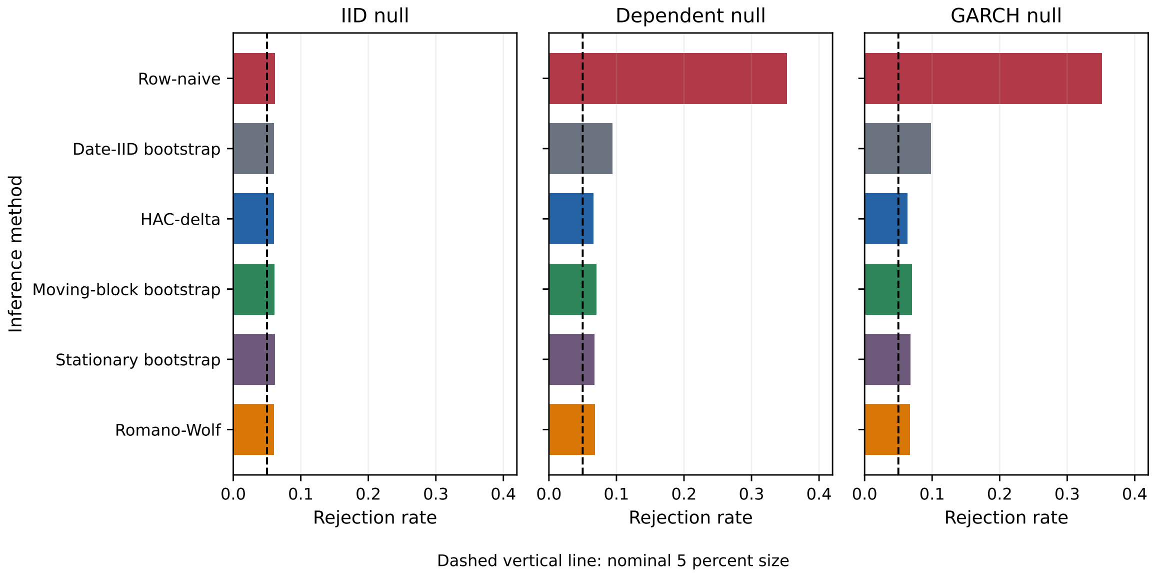

Figure 2. Rejection rates under null designs. The dashed line is the nominal 5 percent rejection rate. Row-naive rejection is near nominal under IID sampling but rises sharply under dependent nulls, while date-level dependence-aware procedures remain much closer to nominal size.

Table 3. Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

9 Discussion and limitations

The results should be read as a financial-econometrics testing exercise for entity-time panel evidence. The framework addresses two common sources of false precision: same-date cross-sectional row multiplication and serial dependence in the date-level return series. It does not remove the need for economic interpretation, out-of-sample validation, liquidity analysis, or venue-specific execution modeling.

The formal validity statement assumes a fixed and finite researcher menu, stationary date-return series, adequate moments for the Sharpe delta-method expansion, and block-bootstrap regularity. The framework does not cover long-memory processes, infinite-variance returns, growing researcher menus, structural breaks, or nonstationary date returns without additional assumptions. The empirical stationarity, holdout, factor-alpha, placebo, and cost outputs are therefore diagnostics and boundary checks, not proofs that every maintained assumption holds in every market setting. A locked holdout period or independently refreshed sample after the researcher menu is frozen remains the stronger design when such data are available.

The same adjustment that controls false positives also reduces apparent power because the effective evidence is closer to the number of active dates than to the number of selected rows. A non-rejection after date-level HAC or dependent-bootstrap adjustment should therefore be read as insufficient date-level evidence under the maintained design, not as proof that the underlying economic effect is exactly zero. Longer date histories, independent holdouts, or stronger prior restrictions on the researcher menu are needed when power is the binding concern.

10 Data and code availability

The reproducibility package contains the LaTeX source, generated public aggregate tables, public-data loaders, simulation code, unit tests, release scripts, and SHA256 provenance. The public repository is: `https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence`

Code is released under Apache-2.0 and manuscript/public aggregate tables under CC-BY 4.0 unless a target venue requires different terms. Raw third-party data are not redistributed where source terms do not permit redistribution; the repository instead provides documented loaders, source links, access metadata, generated aggregate tables, and checksums needed to reproduce the analysis. Proprietary predictions, raw trades, private tickers, production feature definitions, and unrelated workspace files are excluded. The technical appendix contains the full simulation grid, robustness tables, positive-control output, configuration notes, and proof details omitted from the main manuscript for length. The release was checked under Python 3.14.5; package requirements and reproduction instructions are recorded in the release manifest and `requirements-paper.txt`.

11 Conclusion

The paper's central message is simple. In financial panels, multiplying same-date rows does not automatically create independent Sharpe-ratio evidence when the relevant payoff is a date-level portfolio return. The solution proposed here is not a new anomaly but a sampling-boundary method: aggregate to the date-return series, estimate long-run Sharpe uncertainty at that boundary, and record how far the row-pooled report differed from the confirmatory date-level claim.

In public benchmark evidence, this framework preserves the canonical French WML result while overturning weaker stress-panel claims once dependence, permutation validity, researcher menus, and turnover costs are respected. Pre-aggregated AQR factor series remain useful time-series benchmarks, but without row-level constituent signals they cannot be evaluated by the panel permutation check.

The broader implication is that Sharpe-ratio evidence in panels should be judged at the date-level portfolio return boundary. Future work can extend the framework to richer implementation-cost models, time-varying cross-sectional dependence, variable selected-count asymptotics, larger anomaly panels, and richer blocked-permutation designs; those extensions are beyond the scope of this paper.

Author contributions

Dhananjay S. Chauhan is the sole author of this manuscript and was responsible for conceptualization, methodology, software, validation, formal analysis, data curation, writing, and revision.

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Conflicts of interest

The author declares no conflicts of interest.

Ethics statement

This study uses publicly available aggregate financial data and simulation data. It does not involve human participants, personal data, clinical data, animal subjects, or cell lines. Ethics approval was therefore not required.

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